

## Perspective

## Research Direction toward Theoretical Efficiency in Perovskite Solar Cell

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**Research Direction toward Theoretical Efficiency in Perovskite Solar Cell**

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**Abstract**

The recently certified efficiency of 22.7% makes perovskite solar cell (PSC) rise to the top among the thin film technologies of the photovoltaics. The research activities of PSC has been triggered by the groundbreaking report on a 9.7% efficient and 500 h-stable solid-state perovskite solar cell employing methylammonium lead iodide adsorbed on mesoporous TiO<sub>2</sub> film and organic hole conducting layer in 2012. However, PSC is facing issues on stability, current-voltage hysteresis, ion migration, and so on, which should be solved for commercialization. In addition, further improvement in power conversion efficiency (PCE) is waiting for PSC. In this perspective, Shockley–Queisser (S-Q) limit in PSC is investigated, where the best performing state-of-the-art PSC is used for this study. Short-circuit photocurrent density ( $J_{sc}$ ) is found to approach S-Q limit, while open-circuit voltage ( $V_{oc}$ ) and fill factor (FF) are far below their S-Q limits. Thus toward S-Q limit efficiency of ~30% for PSC with light absorber having band gap of 1.6 eV, strategy of reducing non-radiative recombination and interface recombination to achieve theoretical  $V_{oc}$  and FF is more important than finding a method to improve  $J_{sc}$ . For this end, types of defects should be sophisticatedly characterized and engineered although organic-inorganic halide perovskite is known to be defect-tolerant and benign grain boundary.

**Keywords:** perovskite solar cell; Shockley–Queisser; theoretical efficiency; recombination; defect; carrier management

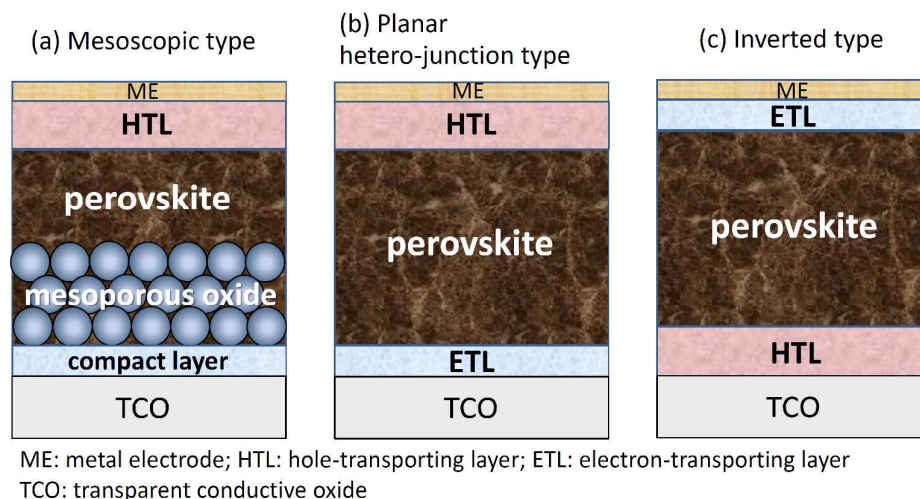
## History and Research Activity of Perovskite Solar Cell

Perovskite solar cell (PSC) is named after breakthrough discoveries of methylammonium lead iodide ( $\text{MAPbI}_3$ , where  $\text{MA} = \text{CH}_3\text{NH}_3$ ) light harvester having perovskite crystal structure that was used as sensitizer in liquid junction type in 2009 [1] and 2011 [2] and in solid-state junction type [3, 4]. Research activity on PSC was triggered by the report on the 9.7% and 500 h-stable solid-state version [3] rather than the two previous reports on perovskite-sensitized solar cells since  $\text{MAPbI}_3$  is rapidly dissolved in liquid electrolyte. Over the years, PSC appears to be a new paradigm in the field of photovoltaics [5, 6]. As a result, power conversion efficiency (PCE) of PSC reached 22.7% that was certified in 2017 [7]. As compared to the conventional thin film solar cells, the PCE of PSC is higher than CIGS (22.6%), CdTe (22.1%) and poly Si (22.3%). Low absorption coefficient molecular dye has limited further improvement of PCE in dye-sensitized solar cell, which motivated scientists to develop new light absorbers such as organic-inorganic halide perovskite [8]. Absorption coefficient of  $\text{MAPbI}_3$  is one order of magnitude higher than that of ruthenium based dye [2], which is one of reasons why perovskite solar cell exhibits superb photovoltaic performance.

As mentioned, a swift surge of perovskite photovoltaics is followed by the report on the stable solid-state perovskite solar cell [2]. Number of publications on perovskite solar cell reaches over 7,000 from 2009 to as of January 5, 2018. Publications per year was obtained using keyword of “perovskite solar” from Web of Science. In 2009, only one paper [1] came out and no paper appeared next year. Although second paper [2] published in 2011 demonstrated improved efficiency, no further works were continued due to instability issue of perovskite in electrolyte. The first report on stable solid-state perovskite solar cell in 2012 actually opened new horizons in perovskite photovoltaics. As a result, publications increased significantly and exponentially. In 2017, more than 3,000 peer-reviewed research articles were published, which was 1.4 times higher than the numbers published in 2016 and 2.3 times higher than those published in 2015. Materials with perovskite structure have been studied not only for solar cell but also for diverse applications such as colossal

magnetoresistance associated with a ferromagnetic-to-paramagnetic phase transition [9], resistive switching memory [10], superconductor [11], or piezoelectricity [12]. Articles on perovskite (keyword of perovskite was used) increased steadily from 747 to 2318 from 1995 to 2012, which results in annual growth rate of about 92 articles/year. On the other hand, number of publications increases from 2625 to 6302 during the period from 2013 to 2017, corresponding to annual growth rate is about 919 articles/year. The year 2013 is inflection point in perovskite-related research, which is obviously due to tremendous works on perovskite solar cells.

The perovskite solar cells can be classified in following three kinds of categories; mesoscopic type, planar hetero-junction type, and inverted type as shown in **Figure 1**. Among the mesoscopic cells,  $\text{TiO}_2$ ,  $\text{ZnO}$ , and  $\text{SnO}_2$ -based cells used conducting scaffold, while  $\text{Al}_2\text{O}_3$ ,  $\text{ZrO}_2$ , and  $\text{SiO}_2$ -based cells used non-conducting scaffold and barrier layer. Most of papers reporting efficiency higher than 20% are based on the mesoscopic type. At present, the advantage of mesoscopic structure is regarded as the effective electron extraction based on the large-area contact between the perovskite layer and the conducting layer. However, there is an effect of the materials used in each structure. Most of the electron transport materials in the mesoscopic type devices are metal oxides, whereas the electron transport materials in the inverted type devices are mainly  $\text{C}_{60}$  derivatives. Selection of the hole transport materials may also be regarded. Hole transport materials based on small molecules (spiro-OMeTAD or FDT) afforded higher open-circuit voltage ( $V_{oc}$ ). In the case of inverted type devices, small molecule-based hole transport materials are rarely used. Investigations for the reason why mesoscopic structure affords better performances may bring further improvements in perovskite solar cells.



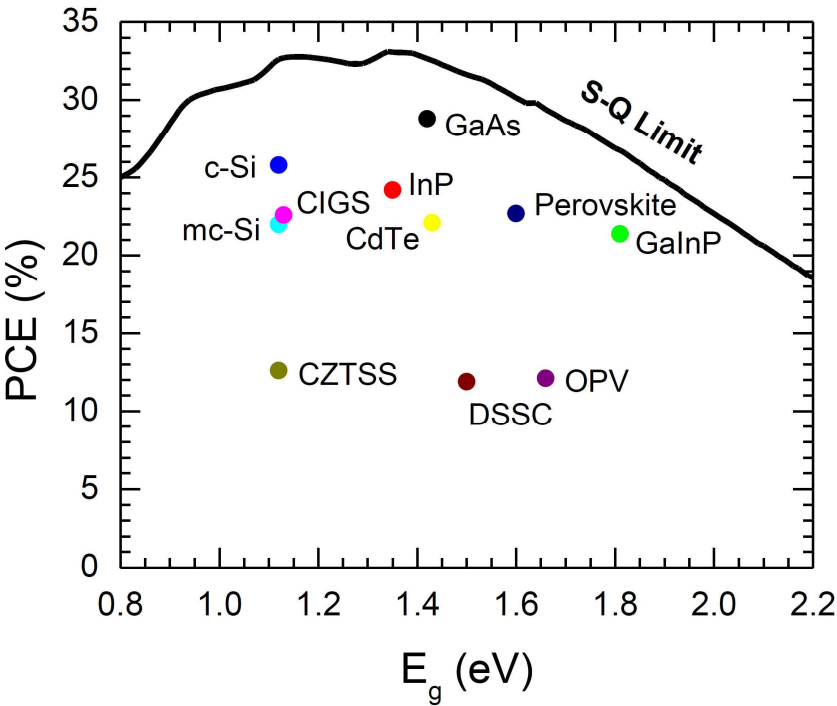
**Figure 1.** Structure of perovskite solar cells: (a) mesoscopic type having mesoporous oxide layer, (b) planar hetero-junction type with an ETL on TCO and a HTL and (c) inverted type with an HTL on TCO and ETL.

## Shockley–Queisser Limit for Perovskite Solar Cell

### Status of record efficiency solar cells relative to Shockley–Queisser limit

The maximum PCE of a single junction solar cell for a given illumination spectrum is known to be Shockley–Queisser limit (S-Q limit) [13], where the only recombination path was assumed to be radiative recombination and photon energy above band gap ( $E_g$ ) was assumed to be converted into electron-hole pair with quantum efficiency of 100%. The sun spectrum was approximated by the emission of a black body with a surface temperature of 6000 K, nearly corresponding to AM 0 with light intensity of  $1576.7 \text{ W/m}^2$ . Recently, the S-Q limit for AM 1.5G ( $1000.4 \text{ W/m}^2$ ) and temperature of 298.15 K was calculated [14], which is displayed in **Figure 2**. The certified record PCEs of laboratory photovoltaic cells, taken from the solar cell efficiency tables version 51 [15], are also plotted in **Figure 2**. The highest PCE of 32.91% (short-circuit photocurrent density ( $J_{sc}$ ) =  $32.88 \text{ mA/cm}^2$ ,  $V_{oc}$  = 1.122 V, and fill factor (FF) = 89.3%) can be expected from solar cell material with  $E_g$  of 1.4 eV. So far, the highest PCE of single junction solar cell was reported to be 28.8% from GaAs with  $E_g$  = 1.42 eV, where  $J_{sc}$  =  $29.68 \text{ mA/cm}^2$ ,  $V_{oc}$  = 1.122 V and FF = 86.5% in which photovoltaic

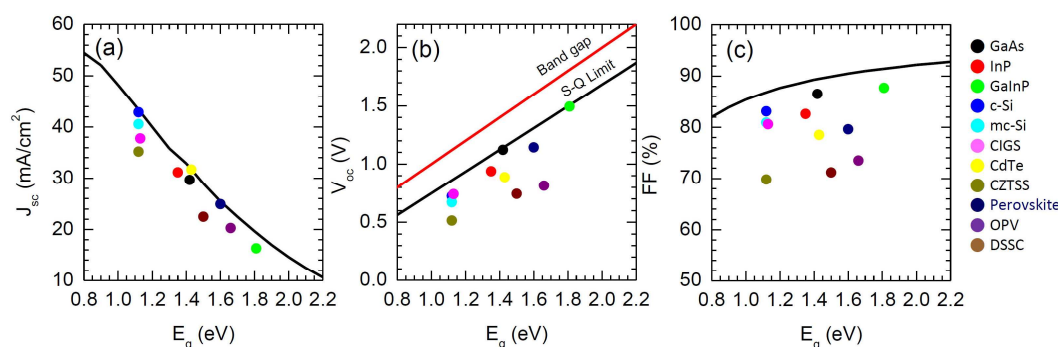
parameters are close to theoretical values. The reported PCE of 28.8% is 87.5% of S-Q limit. For the case of  $E_g = 1.6$  eV corresponding to wavelength of 775 nm, theoretical maximum PCE is calculated to be 30.14% with  $J_{sc} = 25.47$  mA/cm<sup>2</sup>,  $V_{oc} = 1.309$  V and FF = 90.5%. This means that perovskite with  $E_g = 1.6$  eV can reach 30.14%. The best certified PCE of perovskite solar cell was 22.7% [15], where  $J_{sc} = 24.92$  mA/cm<sup>2</sup>,  $V_{oc} = 1.144$  V and FF = 79.6% were reported. The reported PCE of 22.7% is about 75% of S-Q limit, which indicates that there is still large room to improve PCE of perovskite solar cell.



**Figure 2.** The maximum PCE (S-Q limit) for a solar cell operated under AM 1.5G illumination at 298.15 K as a function of the band gap energy ( $E_g$ ). Data points represent record PCEs of laboratory solar cells, which were taken from ref. [15]. Data points taken/adapted from Ref. 15. Copyright John Wiley and Sons.

In **Figure 3**, photovoltaic parameters for record efficiency solar cells are plotted as a function of  $E_g$ .  $J_{sc}$ s of crystalline Si (c-Si), multicrystalline Si (mc-Si) CdTe, GaAs and perovskite are approaching theoretical S-Q limit, while  $V_{oc}$ s are lower than S-Q limit except for GaAs and

GaInP.  $V_{SQ}$  ( $V_{oc}$  for S-Q limit) is 1.122 V, 1.215 V and 1.309 V for  $E_g = 1.4$  eV, 1.5 eV and 1.6 eV, respectively. For the record efficiency perovskite solar cell,  $V_{oc}$  of 1.144 V is 87.4% of S-Q limit and 0.456 V lower than  $E_g$ .  $V_{oc}$ s of CZTSS ( $Cu_2ZnSnS_{4-y}Se_y$ ), OPV (organic photovoltaic cell) and DSSC (dye-sensitized solar cell) fall short of  $V_{SQ}$ . Most of the studied photovoltaic materials show much lower FF than S-Q limit, where GaAs (FF = 86.5%) is close to S-Q limit (89.3%). FF of the record efficiency perovskite solar cell is as high as 79.6% but still far below the theoretical value of 90.5%. Therefore, according to the analysis based on the photovoltaic parameters relative to S-Q limit, managing FF and  $V_{oc}$  in perovskite solar cell is proposed to be important strategy toward S-Q limit of efficiency.

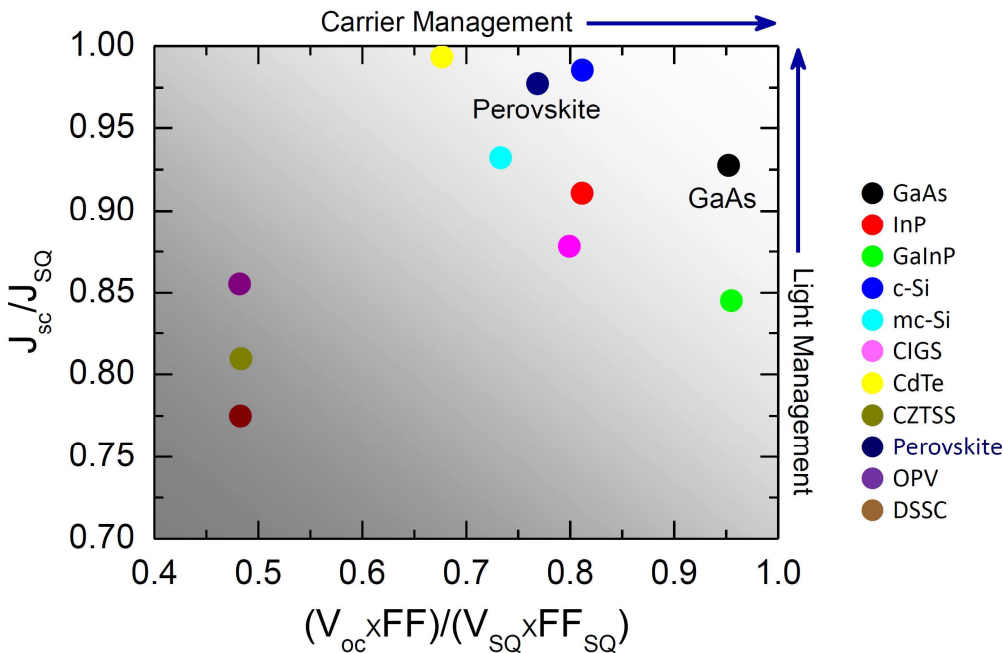


**Figure 3.** Photovoltaic parameters of (a)  $J_{sc}$ , (b)  $V_{oc}$  and (c) FF for record efficiency solar cells with respect to S-Q limit (black line). All data were taken from solar cell efficiency table (version 51) [15]. Data points taken/adapted from Ref. 15. Copyright John Wiley and Sons.

## 2.2 Strategy toward Shockley–Queisser limit

$J_{sc}$  and the  $V_{oc} \times FF$  product of the real devices with respect to the S-Q limiting values were suggested to enable a direct identification of each material in terms of unoptimized light management ( $J_{sc}/J_{SQ} < 1$ ) or carrier management ( $V_{oc} \times FF/V_{SQ} \times FF_{SQ} < 1$ ) [16], where  $J_{SQ}$ ,  $V_{SQ}$  and  $FF_{SQ}$  represent S-Q limiting  $J_{sc}$ ,  $V_{oc}$  and FF values, respectively. Since the data in reference 16 were based on the solar cell efficiency tables (version 45) published in 2015 [17], we updated the data published in the latest version (version 51). **Figure 4** shows the current ratio ( $J_{sc}/J_{SQ}$ ) versus the product of  $V_{oc}$  and FF fractions ( $FF \times V_{oc}/FF_{SQ} \times V_{SQ}$ ) for the record efficiency solar cells. For instance  $J_{sc}/J_{SQ} = 1$  means current density is equal to S-Q limit

current. If this value is less than 1, light management (light absorption) is critical to improve PCE. In case that  $FF \times V_{oc}$  is lower than  $FF_{SQ} \times V_{SQ}$ , carrier management, associated with reducing non-radiative recombination, become more important to increase PCE. **Figure 4** clearly provides the research direction toward S-Q limit, where reducing non-radiative carrier recombination is more than increasing absorption. Solar cells with  $V_{oc}$  or FF lower than the S-Q limit might have problems of bulk or interfacial carrier recombination, series or shunt resistance, or other electrical nonidealities. Thus, issues on recombination (bulk and/or interface) and parasitic resistance will be critical to improve  $V_{oc}$  and FF of perovskite solar cell.



**Figure 4.** The current ratio  $J_{sc}/J_{SQ}$  versus the product of  $V_{oc}$  and FF fractions ( $FF \times V_{oc}/FF_{SQ} \times V_{SQ}$ ) for the record efficiency solar cells. The data were taken from solar cell efficiency tables (version 51) in ref. [15]. Data points taken/adapted from Ref. 15. Copyright John Wiley and Sons.

## Methodologies toward PCE of ~30% using perovskite with $E_g$ of ~1.6 eV

### Understanding carrier mobility and diffusion length

The studied perovskite materials for PSC are classified into pure and mixed perovskite. In  $\text{APbX}_3$  ( $X = \text{I}, \text{Br}, \text{or Cl}$ ) chemical formula, MA ( $= \text{CH}_3\text{NH}_3^+$ ), FA ( $= \text{HC}(\text{NH}_2)_2^+$ ), and Cs can be stabilized in A site in terms of tolerance factor ranging from 0.8 to 1.0. Pure perovskite stands for  $\text{MAPbI}_3$  or  $\text{FAPbI}_3$  and mixed perovskite represents typically mixed cation in A site. Thus (MA, FA), (MA, Cs), (FA, Cs) and (FA, MA, Cs) combinations can be possible. Mixed halide at given cation combination is also possible, for instance,  $\text{I}_{3-x}\text{Br}_x$  or  $\text{Br}_{3-x}\text{Cl}_x$  is available. Sometimes  $\text{MAPbI}_{3-x}\text{Cl}_x$  is described as chemical formula because trace of chloride was detected in the final film [17], which however may not be correct in terms of local structural point of view. In case that chloride is located in axial or equatorial position of  $\text{PbI}_6$  octahedra, lattice constant should be significantly reduced. However, lattice constant is found to be hardly changed, which indicates that the trace chloride might coexist with  $\text{MAPbI}_3$ . Band gap energy of 1.45 ~ 1.6 eV can be easily modified by managing A cation and halide anion in  $\text{APbX}_3$ . Although  $E_g$  is sensitive to halide anion, it can be slightly tuned by A site cation at given anion composition. Pure FA reduces  $E_g$  relative to pure MA.

Understanding carrier (electron and hole) diffusion length will be important in order to carrier management in perovskite solar cell, which was reported in detail with  $\text{MAPbI}_3$  single crystal (MSC) and poly crystal (MPC) thin film [18]. The carrier diffusion length  $L_D$  can be determined using carrier mobility ( $\mu$ ) and carrier life time ( $\tau_r$ ) because of  $L_D = (k_B T \mu \tau_r / e)^{1/2}$  where  $k_B$ ,  $T$  and  $e$  are Boltzman constant, absolute temperature and elementary charge, respectively. Carrier mobility along with trap density can be obtained from dark current measurement. From the hole-only device (Au/MSC/Au or Au/MPC/PEDOT:PSS), hole trap density of MSC was estimated to be  $3.6 \times 10^{10}/\text{cm}^3$ , while that of MPC thin film was  $2.0 \times 10^{15}/\text{cm}^3$ . Hall and SCLC measurements resulted in hole mobility ( $\mu_p$ ) of MSC ranging between 105 and 164  $\text{cm}^2/\text{Vs}$ . Using the electron-only device (Ga/PCBM/MSC/PCBM/Ga), similar electron trap density of  $4.5 \times 10^{10}/\text{cm}^3$  was estimated, while lower electron mobility of 24.8  $\text{cm}^2/\text{Vs}$  was obtained for MSC.  $\tau_r$  of 82-95  $\mu\text{s}$  for MSC was evaluated from transient photovoltaic and impedance spectroscopy. Therefore hole diffusion length was evaluated to be 175  $\mu\text{m}$  for MSC. Although carrier mobility and diffusion length are higher for hole in  $\text{MAPbI}_3$  single crystal, their values are different depending on preparation condition [19]. Balanced electron diffusion length of 130 nm and hole diffusion length of 110 nm were

reported for solution-processed MAPbI<sub>3</sub> thin film in 2013 [20]. However, MAPbI<sub>3</sub> polycrystalline thin films prepared by advanced methods were reported to show diffusion length ranging from ~1 μm to over 20 μm [21-27]. Carrier diffusion length of MAPbI<sub>3</sub> with grain size of 250 nm was estimated to be about 1 μm that is four times larger than the grain size [28]. Recently, 400 μm-thick MPC film having average grain size of 30 μm was developed, where hole mobility was determined to be  $139 \pm 14$  cm<sup>2</sup>/Vs and the electron mobility was  $66 \pm 7$  cm<sup>2</sup>/Vs from time-of-flight method [29], which are almost identical with or even better than the values observed for MSC.

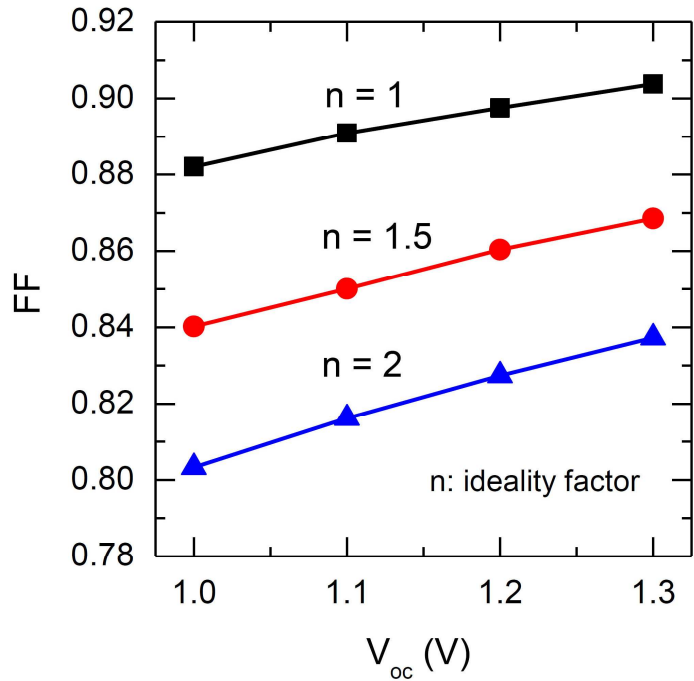
FAPbI<sub>3</sub> is more promising candidate approaching S-Q limit because of lower E<sub>g</sub> of about 1.47 eV [30], at which S-Q limit PCE of about 32% is expected. FAPbI<sub>3</sub> might be optically better than MAPbI<sub>3</sub> because extinction coefficient of FAPbI<sub>3</sub> is larger than that of MAPbI<sub>3</sub> along with small refractive index [31]. Carrier property of FAPbI<sub>3</sub> single crystal (FSC) was studied [32], where time-dependent photoluminescence study revealed that carrier life time of single crystal FAPbI<sub>3</sub> was 484 ns that was longer than that of FAPbI<sub>3</sub> poly crystal (FPC) thin film (227 ns). Carrier mobility of FSC was ranging from 1.07 cm<sup>2</sup>/Vs (TOF method) to 4.4 cm<sup>2</sup>/Vs (SCLC method). Trap density was  $6.2 \times 10^{11}$ /cm<sup>3</sup> that is one order of magnitude lower than MSC, while conductivity of FSC ( $\sigma = 1.1 \times 10^{-7}$ /Ω cm) is one order of magnitude higher than that of MSC due to higher carrier concentration. Bulk diffusion length of FSC was underestimated to be 2.2 μm. Carrier diffusion length of FPC thin film, not pure iodide but mixed iodide and bromide, was reported to be 3.1 μm [33]. For FSC with E<sub>g</sub> of 1.41 eV (5.63 eV and 4.22 eV for valence band maximum and conduction band minimum, respectively), different results on electronic properties were reported [34], where conductivity of  $\sigma = 2.2 \times 10^{-8}$ /Ω cm is lower than the results in reference 29 and trap density of  $1.13 \times 10^{10}$ /cm<sup>3</sup> is higher than reference 32. In addition, longer hole diffusion length of 6.6 μm was estimated. As compared to the balanced electron and hole diffusion length in MAPbI<sub>3</sub>, it is noted that hole diffusion length in FAPbI<sub>3</sub> was ~813 nm that is much longer than electron diffusion length (~117 nm) [35]. In **Table 1**, carrier properties are listed for MSC, MPC, FSC and FPC.

**Table 1.** Carrier mobility, life time and diffusion length for single crystal and poly crystal MAPbI<sub>3</sub> and FAPbI<sub>3</sub>. Parentheses represent reference number.

| Composition                            | Electron mobility ( $\mu_n$ ) | Hole mobility ( $\mu_p$ )          | Electron life time | Hole life time     | Electron diffusion length | Hole diffusion length |
|----------------------------------------|-------------------------------|------------------------------------|--------------------|--------------------|---------------------------|-----------------------|
| MAPbI <sub>3</sub> (single crystal)    | 24.8 cm <sup>2</sup> /Vs [18] | 105 ~ 164 cm <sup>2</sup> /Vs [18] |                    | 82-95 $\mu$ s [18] |                           | 175 $\mu$ m [18]      |
| MAPbI <sub>3</sub> (poly crystal film) |                               |                                    |                    |                    | 130 nm [20]               | 110 nm [20]           |
| FAPbI <sub>3</sub> (single crystal)    |                               | 35 cm <sup>2</sup> /Vs [34]        |                    |                    |                           | 6.6 $\mu$ m [34]      |
| FAPbI <sub>3</sub> (thin film)         |                               | 27 cm <sup>2</sup> /Vs [33]        |                    |                    | 177 nm [35]               | 813 nm [35]           |

### How to improve $V_{oc}$ and FF

According to the equation of  $FF = [V - \ln(V+0.72)]/(V + 1)$ , where  $V$  is normalized  $V_{oc}$  and corresponding to  $(q/nk_B T)V_{oc}$  ( $q$ : elementary charge,  $n$ : ideality factor,  $k_B$ : Boltzmann constant and  $T$ : absolute temperature) [36], FF is calculated with respect to  $V_{oc}$  and  $n$ . **Figure 5** shows that FF increases with  $V_{oc}$  at given  $n$  and should increase with decreasing  $n$  at given  $V_{oc}$ . This indicates that FF is sensitive to both  $n$  and  $V_{oc}$ . High  $V_{oc}$  with small  $n$  (approaching ideal diode) is essential for high FF. Thus understanding factors affecting ideality factor and  $V_{oc}$  is important to design high efficiency PSC. Ideality factor is related to recombination, where the ideal diode with  $n = 1$  assumes that recombination occurs only via band-to-band recombination (radiative recombination) or via Shockley-Read-Hall deep (SRH) recombination (trap-assisted recombination). However, since recombination can occur in other ways,  $n$  is deviated from 1. When considering  $V_{oc}$  is also related to recombination, better understanding recombination in halide perovskite materials and PSC is important to improve FF and  $V_{oc}$ .



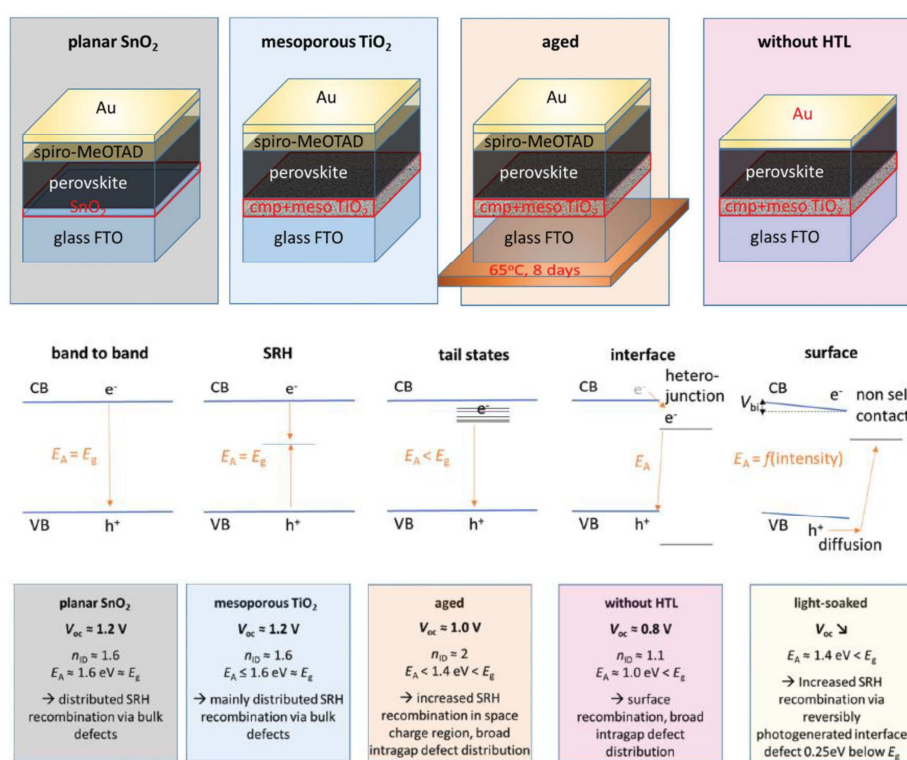
**Figure 5.** Fill factor (FF) as a function of open-circuit voltage ( $V_{oc}$ ) showing that FF increases with  $V_{oc}$ . At given  $V_{oc}$ , FF increases with ideality factor ( $n$ ). The data were calculated using “normalized  $V_{oc}$ ” in refer. [36].

Recently, ideality factors were determined from three different methods, light intensity dependent  $V_{oc}$ , dark current-voltage curve and electroluminescence for planar and mesoscopic PSCs [37], where among the suggested methods  $n$  obtained from light intensity dependent  $V_{oc}$  (see the equation 1) was proved to be more reliable.

$$eV_{oc} = E_g - nk_B T \ln(I_0/I) \quad (1)$$

So far the best performing PSC includes mesoporous  $\text{TiO}_2$  and organic hole conducting material (**Figure 6**). With this device layout  $V_{oc}$  as high as 1.2 V could be obtained and ideality factor was estimated to be  $n = 1.6$ . Planar device layout with thin  $\text{SnO}_2$  layer showed high  $V_{oc}$  of 1.2 V. However,  $V_{oc}$  was declined to 1.0 V upon aging the mesoporous device.  $V_{oc}$  as low as 0.8 V was observed from the device without hole transporting layer (HTL) and a significant reduction of  $V_{oc}$  was detected under light soaking. In **Figure 6**, types of

recombination are shown, where activation energy of recombination,  $E_A$ , estimated from the equation 1 by replacing  $E_g$  with  $E_A$ , are compared with  $E_g$ .  $E_A$  is almost similar to  $E_g$  for the mesoporous  $\text{TiO}_2$ -based PSC, but it is lowered by about 0.2 V upon aging and light soaking. From the studies on ideality factor, it was suggested that an  $n$  between 1.5 and 2 and an  $E_A = E_g$  for the best performing devices is attributed to SRH recombination in the bulk of a mostly intrinsic or depleted perovskite film, while  $E_A < E_g$  case is due to recombination of charge carriers at interfaces of heterojunctions or surfaces [37]. Thus, although organic-inorganic halide perovskite has defect-tolerant [38] and benign grain boundary effects [39], careful control of defects in ppm regime is critical toward higher PCE.

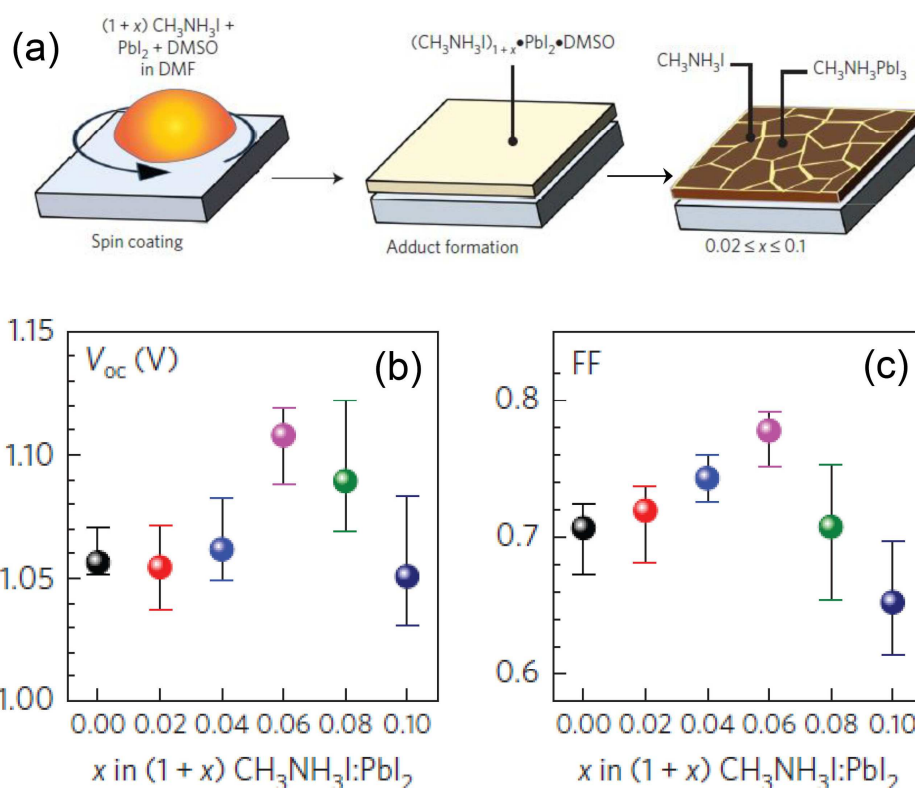


**Figure 6.** (Top) Device layouts for studying ideality factor, (middle) illustration of the activation energy for recombination ( $E_A$ ) depending on the recombination process and (bottom) ideality factor ( $n$ ) and  $E_A$  values for different device layouts along with aged and light-soaked conditions. Reprinted with permission from ref. [37]. Copyright 2018, Royal Society of Chemistry.

### Experimental approaches to improve $V_{oc}$ and FF via defect engineering

In order to accomplish high  $V_{oc}$ , 'mixed cation' has been useful. Grätzel et al. reported the mix-cation perovskite with Rb, Cs, MA, and FA [40]. The system afforded 21.8% PCE with  $V_{oc}$  of 1.18 V, while the fully illuminated device without mask afforded  $V_{oc}$  of 1.24 V. Segawa et al. reported potassium-doped perovskite solar cell with  $V_{oc}$  of 1.16 V and PCE of 20.5% where the I-V hysteresis was minimized [41]. On the other hand, very high FF 0.817 has been reported by the use of poly(triarylamine) as hole-transport layer for the high efficiency cell with 22.6% PCE [42], where the perovskite solar cells were mesoscopic type with conducting scaffold.

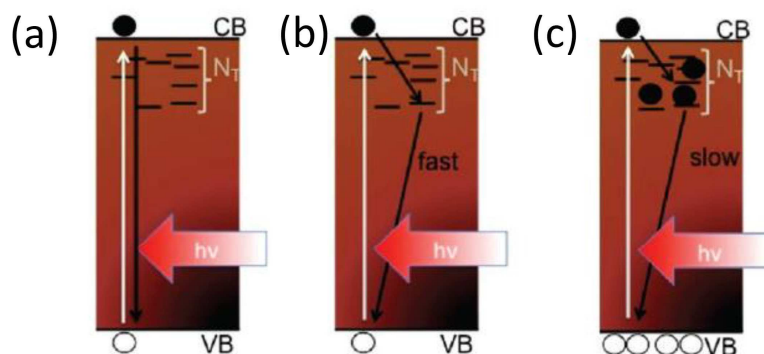
Non-stoichiometric Lewis acid-base adduct method was found to heal grain boundary because excess chemical will be unreacted and placed at grain boundary or surface [43]. Excess chemical was found to be placed at grain boundary as a thin layer when excess MAI was included in precursor solution [44]. In **Figure 7(a)**, non-stoichiometric adduct method is schematically presented, where final perovskite film is formed via adduct. Change in concentration of the excess MAI affects photovoltaic parameters in which  $V_{oc}$  and FF increase as the concentration increases (**Figure 7(b) and (c)**). The 0.06 mol excess MAI results in increase of  $V_{oc}$  from 1.029 V (stoichiometric sample) to 1.118 V, along with FF from 0.683 to 0.755. Time-resolved photoluminescence and transient absorption spectroscopic studies revealed that MAI excess sample ( $x = 0.06$ ) showed well-suppressed non-radiative recombination at grain boundaries and significantly improved carrier life time of bulk perovskite as well. For the non-passivated grain boundaries, the vacancy or interstitial defects at/near grain boundaries was reported to induce detrimental charge trap states resulting in lower quasi Fermi level and reduced carrier mobility, which might reduce  $V_{oc}$  and FF [45-49]. The importance of grain boundary engineering could reduce trap-mediated recombination (kind of SRH recombination) by passivating the structural defect at/near the grain boundaries.



**Figure 7.** (a) Schematic representation of the coating process used to form the CH<sub>3</sub>NH<sub>3</sub>PbI<sub>3</sub> perovskite films with grain boundary healing layer via non-stoichiometric Lewis acid–base adduct intermediates. (b) Plots of V<sub>oc</sub> and FF as a function of x in (1+x)CH<sub>3</sub>NH<sub>3</sub>I : PbI<sub>2</sub>. Reprinted with permission from ref. [44]. Copyright 2016, Springer Nature.

Three types of recombination processes were analyzed to investigate their effect on V<sub>oc</sub> [50], which are trap-filled bimolecular recombination (radiative recombination), fast trap-mediated recombination and slow (long-lived hole) trap-mediated recombination (**Figure 8**). V<sub>oc</sub> was estimated from  $E_g - [(E_{CB} - E_{Fn}) + (E_{VB} - E_{Fp})]$  for the MAPbI<sub>3</sub> with  $E_g = 1.6$  eV, where  $E_g$ ,  $E_{CB}$ ,  $E_{VB}$ ,  $E_{Fn}$  and  $E_{Fp}$  represent band gap, conduction band level, valence band level, quasi-Fermi energy for electron and quasi-Fermi energy level for hole, respectively. V<sub>oc</sub> = 1.14 V, 1.30 V and 1.26 V was determined for fast (100 ns) trap-mediated recombination, bimolecular recombination and slow trap-mediated recombination, respectively. Since the highest V<sub>oc</sub> of about 1.2 V was experimentally observed from MAPbI<sub>3</sub>, slow trap-mediated recombination

process was expected to dominate. The results suggest that removal of trap states can lead to  $V_{oc}$  as high as 1.3 V that is close to  $V_{SQ}$ .



**Figure 8.** Recombination processes: (a) Trap-filled bimolecular recombination, (b) fast (100 ns) trap-mediated recombination and (c) slow (Long-lived holes) trap-mediated recombination. Reprinted with permission from ref. [50]. Copyright 2016, Royal Society of Chemistry.

## Summary

We have investigated photovoltaic parameters of the state-of-the-art PSC with PCE of 22.7% in terms of S-Q limit values. Photocurrent density is already positioned at S-Q limit, while  $V_{oc}$  (1.114 V) is  $\sim 0.165$  V lower than S-Q limit (1.309 V) and FF (79.6%) is much lower than S-Q limit (90.5%). Research in PSC is proposed to manage carriers associated with recombination issue. For this purpose, we propose here that ideal diode employing defect-minimized perovskite material is essential to approach S-Q limit efficiency. Defects in PSC depend on perovskite material composition, grain size, crystallinity, grain boundary and interface at perovskite/ETL (electron transporting layer) or HTL (hole transporting layer). In order to reach theoretical  $V_{oc}$  and FF, ideal diode with allowing only radiative recombination is required. This means SRH recombination and interfacial recombination should be eliminated. According to recent report [51],  $V_{oc}$  of 1.33 V (slightly higher than S-Q value. refer to the reference 51 for further reading) and FF of 0.91 are available when only radiative recombination is presented (PCE will be over 30% for a  $E_g = 1.6$  eV perovskite material), while  $V_{oc}$  and FF are decreased to 1.16 V and 0.86 in case of SRH recombination is involved,

which is further reduced to 0.97 V and 0.8 in the presence of interface recombination. Ideality factor is decreased from 1 to 1.5 in the presence of SRH recombination and further to 2 when adding interface recombination.  $V_{oc}$  value is good indicator to investigate which recombination is presented in your device. This is also good example how recombination affect photovoltaic parameters and PCE. Here are some suggestions how to minimize SRH and interface recombination. Since PSC contains many heterojunctions, care should be taken to make best heterojunction to reduce interface recombination via not only energetics but also physico-chemical interaction between perovskite and ETL or HTL. In addition, since ETL or HTL, depending on PSC type such as normal or inverted, can provide a kind of substrate for growing perovskite film, surface chemistry of ETL or HTL seems to be important to minimize interface recombination and bulk defects as well.

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